ICM 42688 `IMU 0` | `SPI 1`

- User Bank 0 Registers

Addr (Hex)	Addr (Dec)	Register Name	Data
14	20	INT_CONFIG	`0b110`
1D	29	TEMP_DATA1	TEMP_DATA[15:8]
1E	30	TEMP_DATA0	TEMP_DATA[7:0]
1F	31	ACCEL_DATA_X1	ACCEL_DATA_X[15:8]
20	32	ACCEL_DATA_X0	ACCEL_DATA_X[7:0]
21	33	ACCEL_DATA_Y1	ACCEL_DATA_Y[15:8]
22	34	ACCEL_DATA_Y0	ACCEL_DATA_Y[7:0]
23	35	ACCEL_DATA_Z1	ACCEL_DATA_Z[15:8]
24	36	ACCEL_DATA_Z0	ACCEL_DATA_Z[7:0]
25	37	GYRO_DATA_X1	GYRO_DATA_X[15:8]
26	38	GYRO_DATA_X0	GYRO_DATA_X[7:0]
27	39	GYRO_DATA_Y1	GYRO_DATA_Y[15:8]
28	40	GYRO_DATA_Y0	GYRO_DATA_Y[7:0]
29	41	GYRO_DATA_Z1	GYRO_DATA_Z[15:8]
2A	42	GYRO_DATA_Z0	GYRO_DATA_Z[7:0]
2D	45	INT_STATUS	
4B	75	SIGNAL_PATH_RESET	`0b100`
4E	78	PWR_MGMT0	`0b010011`
4F	79	GYRO_CONFIG0	
50	80	ACCEL_CONFIG0	
54	84	TMST_CONFIG	`0b10011`
63	99	INT_CONFIG0	`0b00100000`
65	101	INT_SOURCE0	`0b00011000`
70	112	SELF_TEST_CONFIG	
75	117	WHO_AM_I	
76	118	REG_BANK_SEL	

- User Bank 1 Registers

Addr (Hex)	Addr (Dec)	Register Name	Data
5F	95	XG_ST_DATA	X-gyro self test data
60	96	YG_ST_DATA	Y-gyro self test data
61	97	ZG_ST_DATA	Z-gyro self test data
62	98	TMSTVAL0	TMST_VALUE[7:0]
63	99	TMSTVAL1	TMST_VALUE[15:8]
64	100	TMSTVAL2	TMST_VALUE[19:16]

- User Bank 2 Registers

Addr (Hex)	Addr (Dec)	Register Name	Data
3B	59	XA_ST_DATA	X-accel self test data
3C	60	YA_ST_DATA	Y-accel self test data
3D	61	ZA_ST_DATA	Z-accel self test data

- Characteristics
  - Gyroscope
    - Gyroscope range: 2000 dps
    - Sensitivity scale factor for 2k dps: 16.4 LSB/dps
    - Initial zero rate output tolerance: +- 0.5 dps
    - Gyroscope start up time: 30 ms
    - ODR: 1 kHz
  - Accelerometer
    - Accelerometer range: 16 g
    - Sensitivity scale factor for 16 g: 2048 LSB/g
    - Initial zero rate output tolerance: +- 20 mg
    - Accelerometer start up time: 10 ms
    - ODR: 1 kHz
  - Temperature in Degrees Centigrade = (TEMP_DATA / 132.48) + 25
- SPI
  - Pull CS to activate: LOW
  - Data delivered MSB first LSB last
  - Data transitioned on falling edge of SCLK

- Self test at boot for later version
- Interrupts
    - INT fire pull: LOW
    - Use INT 1
    - INT fired when DATA READY, cleared when DATA READ
- Time stamp of reading stored in register
- Power mode: 6-axis low-noise mode